
UNIT 8 FUZZY AND ROUGH SETS

Structure

- 8.0 Introduction
- 8.1 Objectives
- 8.2 Fuzzy Systems
- 8.3 Introduction to Fuzzy Sets
- 8.4 Fuzzy Set Representation
- 8.5 Fuzzy Reasoning
- 8.6 Fuzzy Inference
- 8.7 Rough Set Theory
- 8.8 Summary
- 8.9 Solutions/ Answers
- 8.10 Further Readings

8.0 INTRODUCTION

In the earlier units, we discussed PL and FOPL systems for making inferences and solving problems requiring logical reasoning. However, these systems assume that the domain of the problems under consideration is complete, precise and consistent. But, in the real world, the knowledge of the problem domains is generally neither precise nor consistent and is hardly complete.

In this unit, we discuss a number of techniques and formal systems that attempt to handle some of these blemishes. To begin with we discuss the **fuzzy systems** that attempt to handle **imprecision** in knowledge bases, specially, due to use of natural language words like hot, good, tall etc.

Then, we discuss **non-monotonic systems** which deal with **indefiniteness** of knowledge in the knowledge bases. The significance of these systems lies in the fact that most of the statements in the knowledge bases are actually based on **beliefs** of the concerned persons or actors. These beliefs get revised as better evidence for some other beliefs become available, where the later beliefs may be in conflict with the earlier beliefs. In such cases, the earlier beliefs may have to be temporarily suspended or permanently excluded from further considerations.

Subsequently, we will discuss two formal systems that attempt to handle **incompleteness** of the available information. These systems are called **Default Reasoning Systems** and **Closed World Assumption Systems**. Finally, we discuss some inference rules, viz, **abductive** inference rule and **inductive** inference rule that are, though not deductive, yet are quite useful in solving problems arising out of everyday experience.

8.1 OBJECTIVES

After going through this unit, you should be able to:

- enumerate various formal methods, which deal with different types of blemishes like incompleteness, imprecision and inconsistency in a knowledge base;
- discuss, why fuzzy systems are required;
- discuss, develop and use fuzzy arithmetic tools in solving problems, the descriptions of which involve imprecision;
- discuss default reasoning as a tool for handling incompleteness of knowledge;
- discuss Closed World Assumption System, as another tool for handling incompleteness of knowledge, and
- discuss and use non-deductive inference rules like abduction and induction, as tools for solving problems from everyday experience.

8.2 FUZZY SYSTEMS

In the symbolic Logic systems like, PL and FOPL, that we have studied so far, any (closed) formula has a truth-value which must be binary, viz., True or False. However, in our everyday experience, we encounter problems, the descriptions of which involve some words, because of which, to statements of situations, it is not possible to assign a truth value: True or False. For example, consider the statement:

If the water is too hot, add normal water to make it comfortable for taking a bath.

In the above statement, for a number of words/phrases including ‘too hot’ ‘add’, ‘comfortable’ etc., it is not possible to tell when exactly water is too hot, when water is (at) normal (temperature), when exactly water is comfortable for taking a bath.

For example, we cannot tell the temperature T such that for water at temperature T or less, truth value False can be associated with the statement ‘Water is too hot’ and at the same time truth-value True can also be associated to the same statement ‘Water is too hot’ when the temperature of the water is, say, at degree $T + 1$, $T + 2$etc.

Some other cases of Fuzziness in a Natural Language

Healthy Person: we cannot even enumerate all the parameters that **determine** health. Further, it is even more difficult to tell for what value of a particular parameter, one is healthy or otherwise.

Old/young person: It is not possible to tell exactly upto exactly what age, one is young and, by just addition of one day to the age, one becomes old. We age gradually. Aging is a continuous process.

Sweet Milk: Add small sugar cube one at a time to glass of milk, and go on adding upto, say, 100 small cubes.

Initially, without sugar, we may take milk as not sweet. However, with addition of each one small sugar particle cube, the sweetness **gradually** increases. It is not possible to say that after addition of 100 small cubes of sugar, the milk becomes sweet, and, till addition of 99 small cubes, it was not sweet.

Pool, Pond, Lake, Sea, Ocean: for different sized water bodies, we can not say when exactly a pool becomes a pond, when exactly a pond becomes a lake and so on.

One of the reasons, for this type of problem of our inability to associate one of the two-truth values to statements describing everyday situations, is due to the use of natural language words like hot, good, beautiful etc. Each of these words does not denote something constant, but is a sort of linguistic variable. The context of a particular usage of such a word may delimit the scope of the word as a linguistic variable. The range of values, in some cases, for some phrases or words, may be very large as can be seen through the following three statements:

- Dinosaurs ruled the earth for a long period (*about millions of years*)
- It has not rained for a long period (*say about six months*).
- I had to wait for the doctor for a long period (*about six hours*).

Fuzzy theory provides means to handle such situations. A **Fuzzy theory** may be thought as a technique of providing ‘**continuization**’ to the otherwise binary disciplines like Set Theory, PL and FOPL.

Further, we explain how using fuzzy concepts and rules, in situation like the ones quoted below, we, the human beings solve problems, despite ambiguity in language.

Let us recall the case of crossing a road discussed in Unit 1 of Block 1. We

Mentioned that a step by step method of crossing a road may consist of

- (i) Knowing (exactly) the distances of various vehicles from the path to be followed to cross over.
- (ii) Knowing the velocities and accelerations of the various vehicles moving on the road within a distance of, say, one kilometer.
- (iii) Using Newton’s Laws of motion and their derivatives like $s = ut + at^2$, and calculating the time that would be taken by each of the various vehicles to reach the path intended to be followed to cross over.
- (iv) Adjusting dynamically our speeds on the path so that no collision takes place with any of the vehicle moving on the road.

But, we know the human beings not only do not follow the above precise method but cannot follow the above precise method. **We, the human beings rather feel comfortable with fuzziness than precision.** We feel comfortable, if the instruction for crossing a road is given as follows:

Look on both your left hand and right hand sides, particularly in the beginning, to your right hand side. If there is no vehicle within reasonable distance, then attempt to cross the road. You may have to retreat back while crossing, from somewhere on the road. Then, try again.

The above instruction has a number of words like left, right (it may 45° to the right or 90° to the right) reasonable, each of which does not have a definite meaning. But we feel more comfortable than the earlier instruction involving precise terms.

Let us consider another example of *our being comfortable with imprecision than precision*. The statement: ‘*The sky is densely clouded*’ is more comprehensible to human beings than the statement: ‘*The cloud cover of the sky is 93.5 %*’.

Thus is because of the fact that, we, the human beings are still better than computers in **qualitative** reasoning. Because of better qualitative reasoning capabilities

- just by looking at the eyes only and/or nose only, we may recognize a person.
- just by taking and feeling a small number of grains from cooking rice bowl, we can **tell** whether the rice is properly cooked or not.
- just by looking at few buildings, we can identify a locality or a city.

Achieving Human Capability

In order that computers achieve human capability in solving such problems, computers must be able to solve problems for which *only incomplete and/or imprecise* information/knowledge is available.

Modelling of Solutions and Data/Information/Knowledge

We know that for any problem, the plan of the proposed solution and the relevant information is fed in the computer in a form acceptable to the computer.

However, the problems to be solved with the help of computers are, in the first place, felt by the human beings. And then, the plan of the solution is also prepared by human beings.

It is conveyed to the computer mainly for execution, because computers have much better executional speed.

Summarizing the discussion, we conclude the following facts

- (i) We, the human beings, sense problems, desire the problems to be solved and express the problems and the plan of a solution using imprecise words of a natural language.
- (ii) We use computers to solve the problems, because of their executional power.
- (iii) Computers function better, when the information is given to the computer in terms of *mathematical entities* like numbers, sets, relations, functions, vectors, matrices graphs, arrays, trees, records, etc., and when the *steps of solution* are generally precise, involving no ambiguity.

In order to meet the mutually conflicting requirements:

- (i) *Imprecision* of natural language, with which the human beings are comfortable, where human beings feel a problem and plan its solution.
- (ii) *Precision* of a formal system, with which computers operate efficiently, where computers execute the solution, generally planned by human beings a new formal system viz. Fuzzy system based on the concept of ‘Fuzzy’ was suggested for the first time in 1965 by L. Zadeh.

In order to initiate the study of Fuzzy systems, we quote two statements to recall the difference between a precise statement and an imprecise statement.

A precise Statement is of the form: ‘If income is more than 2.5 lakhs then tax is 10% of the taxable income’.

An *imprecise* statement may be of the form: ‘If the *forecast* about the rain being *slightly less* than previous year is *believed*, then there is around 30% **probability** that economy may suffer heavily’.

The **concept of ‘Fuzzy’**, which when applied as a **prefix/adjective to mathematical entities like set, relation, functions, tree, etc., helps us in modelling the imprecise data, information or knowledge through mathematical tools.**

Crisp Set/Relation vs. Fuzzy Set/Relation: In order to differentiate the sets, normally used so far, from the fuzzy sets to be introduced soon, we may call the normally called sets as crisp sets.

Next, we explain, how the fuzzy sets are defined, using mathematical entities, **to capture imprecise concepts**, through an example of the concept : tall.

In Indian context, we may say, a *male adult*, is

- (i) **definitely tall** if his height > 6 feet
- (ii) **not at all tall** if height < 5 feet and
- (iii) if his height = 5' 2" **a little bit tall**
- (iv) if his height = 5' 6" **slightly tall**
- (v) if height = 5' 9" **reasonably tall** etc.

Next step is **to model ‘definitely tall’ ‘not at all tall’, ‘little bit tall’, ‘slightly tall’ ‘reasonably Tall’** etc. in terms of mathematical entities, e.g., numbers; sets etc.

In **modelling the vague concept like ‘tall’, through fuzzy sets**, the numbers in the **closed set $[0, 1]$ of reals** may be used on the following lines:

- (i) **‘Definitely tall’** may be represented as **‘tallness having value 1’**
- (ii) **‘Not at all tall’** may be represented as **‘Tallness having value 0’**

other adjectives/adverbs may have values between 0 and 1 as follows:

(iii) '*A little bit tall*' may be represented as '*tallness having value say .2*'.

(iv) '*Slightly tall*' may be represented as '*tallness having value say .4*'.

(v) '*Reasonably tall*' may be represented as '*tallness having value say .7*'.

and so on.

Similarly, the values of other concepts or, rather, other **linguistic variables like sweet, good, beautiful, etc. may be considered in terms of real numbers between 0 and 1.**

Coming back to the *imprecise concept of tall*, let us think of five male persons of an organisation, viz., Mohan, Sohan, John, Abdul, Abrahm, with heights 5' 2", 6' 4",

5' 9", 4' 8", 5' 6" respectively.

Then had we talked only of crisp set of tall persons, we would have denoted the

Set of tall persons in the organisation = {Sohan}

But, a fuzzy set, representing tall persons, include *all the persons alongwith respective degrees of tallness*. Thus, *in terms of fuzzy sets*, we write:

Tall = {Mohan/.2; Sohan/1; John/.7; Abdul/0; Abrahm/.4}.

The values .2, 1, .7, 0, .4 are called *membership values or degrees*:

Note: *Those elements which have value 0 may be dropped e.g.*

Tall may also be written as Tall = {Mohan/.2; Sohan/1; John/.7; Abrahm/.4}, neglecting Abdul, with associated degree zero.

If we *define short/Diminutive* as exactly *opposite of Tall* we may say

Short = {Mohan/.8; Sohan/0; John/.3; Abdul/1; Abrahm/.6}

8.3 INTRODUCTION TO FUZZY SETS

In the case of Crisp sets, we have the concepts of Equality of sets, Subset of a set, and Member of a set, as illustrated by the following examples:

(i) Equality of two sets

Let $A = \{1, 4, 3, 5\}$

$B = \{4, 1, 3, 5\}$

$C = \{1, 4, 2, 5\}$

be three given sets.

Then, Set A is equal to set B denoted by $A = B$. But A is not equal to C, denoted by

$A \neq C$.

(ii) **Subset**

Consider sets $A = \{1, 2, 3, 4, 5, 6, 7\}$

$B = \{4, 1, 3, 5\}$

$C = \{4, 8\}$

Then B is a subset of A, denoted by $B \subset A$. Also C is not a subset of A, denoted by $C \not\subset A$.

(iii) **Belongs to/is a member of**

If $A = \{1, 4, 3, 5\}$

Then each of 1, 4, 3 and 5 is called an element or member of A and the fact that 1 is a member of A is denoted by $1 \in A$.

Corresponding Definitions/ concepts for Fuzzy Sets

In order to define for fuzzy sets, the concepts corresponding to the concepts of *Equality of Sets*, *Subset* and *Membership of a Set* considered so far only for crisp sets, first we illustrate the concepts through an example:

Let X be the set on which fuzzy sets are to be defined, e.g.,

$X = \{\text{Mohan, Sohan, John, Abdul, Abraham}\}$.

Then X is called the Universal Set.

Note: *In every fuzzy set, all the elements of X with their corresponding memberships values from 0 to 1, appear.*

(i) Degree of Membership: In respect of fuzzy sets, we do not speak of just 'membership', but speak of 'degree of membership'.

In the set

$A = \{\text{Mohan}/.2; \text{Sohan}/1; \text{John}/.7; \text{Abrahm}/.4\}$,

Degree (Mohan) = .2, degree (John) = .4

For (ii) Equality of Fuzzy sets: Let A, B and C be fuzzy sets defined on X as follows:

Let $A = \{\text{Mohan}/.2; \text{Sohan}/1; \text{John}/.7; \text{Abrahm}/.4\}$

$B = \{\text{Abrahm}/.4, \text{Mohan}/.2; \text{Sohan}/1; \text{John}/.7\}$.

Then, as degrees of each element in the two sets, equal; we say fuzzy set A equals fuzzy set B, denoted as $A = B$

However, if $C = \{\text{Abrahm}/.2, \text{Mohan}/.4; \text{Sohan}/1; \text{John}/.7\}$, then

$A \neq C$.

(iii) **Subset/Superset**

Intuitively, we know

- (i) The **Set of ‘Very Tall’ people** should be a **subset** of the set of **Tall people**.
- (ii) If the **degree of ‘tallness’** of a person is **say .5** then degree of **‘Very Tallness’** for the person should be **lesser say .3**.

Combining the above two ideas we, may say that if

$A = \{\text{Mohan}/.2; \text{Sohan}/1; \text{John}/.7; \text{Abrahm}/.4\}$ and

$B = \{\text{Mohan}/.2, \text{Sohan}/.9, \text{John}/.6, \text{Abraham}/.4\}$ and further,

$C = \{\text{Mohan}/.3, \text{Sohan}/.9, \text{John}/.5, \text{Abraham}/.4\}$,

then, in view of the fact that for each element, degree in A is greater than or equal to degree in B, B is a subset of A denoted as $B \subset A$.

However, degree (Mohan) = .3 in C and degree (Mohan) = .2 in A,

,therefore, C is **not** a subset of A.

On the other hand degree (John) = .5 in C and degree (John) = .7 in A,

therefore, A is also not a subset of C.

We generalize the ideas illustrated through examples above

Let **A and B be fuzzy sets** on the universal set $X = \{x_1, x_2, \dots, x_n\}$ (**X is called the Universe or Universal set**) such that

$A = \{x_1/v_1, x_2/v_2, \dots, x_n/v_n\}$ and $B = \{x_1/w_1, x_2/w_2, \dots, x_n/w_n\}$

with that $0 \leq v_i, w_i \leq 1$. Then fuzzy set A equals fuzzy set B, denoted as $A = B$, if and only if $v_i = w_i$ for all $i = 1, 2, \dots, n$. Further if $w_i \leq v_i$ for all i . then B is a fuzzy subset of A.

Example: Let $X = \{\text{Mohan}, \text{Sohan}, \text{John}, \text{Abdul}, \text{Abrahm}\}$

$A = \{\text{Mohan}/.2; \text{Sohan}/1; \text{John}/.7; \text{Abrahm}/.4\}$

$B = \{\text{Mohan}/.2, \text{Sohan}/.9, \text{John}/.6, \text{Abraham}/.4\}$

Then B is a fuzzy subset of A.

In respect of fuzzy sets vis-à-vis (crisp) sets, we may note that:

- ◆ Corresponding to the concept of ‘belongs to’ of (Crisp) set, we use the concept of ‘degree of membership’ for fuzzy sets.
- ◆ It may be noted that every crisp set may be thought of as a Fuzzy Set, but not conversely. For example, if Universal set is

$X = \{\text{Mohan}, \text{Sohan}, \text{John}, \text{Abdul}, \text{Abrahm}\}$ and

A = set of those members of X who are at least graduates, say,

$= \{\text{Mohan}, \text{John}, \text{Abdul}\}$

then we can rewrite A as a fuzzy set as follows:

$A = \{\text{Mohan}/1; \text{Sohan}/0; \text{John}/1; \text{Abdul}/1; \text{Abrahm}/0\}$, in which degree of each member of the crisp set, is taken as one and degree of each element of the universal set which does not appear in the set A , is taken as zero.

However, conversely, a fuzzy set may not be written as a crisp set. Let C be a fuzzy set denoting **Educated People**, where **degree of education is defined** as follows:

degree of education (Ph.D. holders) = 1

degree of education (Masters degree holders) = 0.85

degree of education (Bachelors degree holders) = .6

degree of education (10 + 2 level) = 0.4

degree of education (8th Standard) = 0.1

degree of education (less than 8th) = 0.

Let us $C = \{\text{Mohan}/.85; \text{Sohan}/.4; \text{John}/.6; \text{Abdul}/1; \text{Abrahm}/0\}$.

Then, we cannot think of C as a crisp set.

Next, we define some more concepts in respect of fuzzy sets.

Definition: Support set of a Fuzzy Set, say C , is a crisp set, say D , containing all the elements of the universe X for which degree of membership in Fuzzy set is **positive. Let us consider again**

$C = \{\text{Mohan}/.85; \text{Sohan}/.4; \text{John}/.6; \text{Abdul}/1; \text{Abrahm}/0\}$.

Support of $C = D = \{\text{Mohan}, \text{Sohan}, \text{John}, \text{Abdul}\}$, where **the element Abrahm does not belong to D , because, it has degree 0 in C .**

Definition: Fuzzy Singleton is a fuzzy set in which there is exactly one element which has positive membership value.

Example:

Let us define a fuzzy set **OLD** on universal set X in which degree of **OLD** is zero if a person in X is below 20 years and Degree of **Old** is .2 if a person is between 20 and 25 years and further suppose that

$\text{Old} = C = \{\text{Mohan}/0; \text{Sohan}/0; \text{John}/.2; \text{Abdul}/0; \text{Abrahm}/0\}$,

then support of **old** = **{John}** and hence **old** is a fuzzy singleton.

Check Your Progress - 1

Ex. 1: Discuss equality and subset relationship for the following fuzzy sets defined on the Universal set $X = \{a, b, c, d, e\}$

$A = \{a/.3, b/.6, c/.4, d/0, e/.7\}$

$B = \{a/.4, b/.8, c/.9, d/.4, e/.7\}$

$C = \{a/.3, b/.7, c/.3, d/.2, e/.6\}$

8.4 FUZZY SET REPRESENTATION

For **Crisp sets**, we have the operations of **Union, intersection & complementation**, as illustrated by the example:

Let $X = \{x_1, x_2, \dots, x_{10}\}$

$A = \{x_2, x_3, x_4, x_5\}$

$B = \{x_1, x_3, x_5, x_7, x_9\}$

Then $A \cup B = \{x_1, x_2, x_3, x_4, x_5, x_7, x_9\}$

$A \cap B = \{x_3, x_5\}$

$A' \text{ or } X \sim A = \{x_1, x_6, x_7, x_8, x_9, x_{10}\}$

The concepts of Union, intersection and complementation for **crisp sets may be extended to FUZZY sets** after observing that for crisp sets A and B, we have

- (i) $A \cup B$ is the **smallest** subset of X **containing** both A and B.
- (ii) $A \cap B$ is the **largest** subset of X **contained in** both A and B.
- (iii) **The complement A' is such that**
 - (a) A and A' **do not have any element in common** and
 - (b) Every element of the universal set is in either A or A'.

Fuzzy Union, Intersection, Complementation:

In order to motivate proper definitions of these operations, we may recall

- (1) when a **crisp set** is treated as a **fuzzy set** then
 - (i) membership in a crisp set is indicated by degree/value of membership as 1 (one) in the corresponding Fuzzy set,
 - (ii) non-membership of a crisp set is indicated by degree/value of membership as zero in the corresponding Fuzzy Set.

Thus, **smaller the value** of degree of membership, a sort of **lesser it is a member** of the Fuzzy set.

(2) While taking union of Crisp sets, members of both sets are included, and none else. However, in each Fuzzy set, all members of the universal set occur but their degrees determine the level of membership in the fuzzy set.

The facts under (1) and (2) above, lead us to define:

The Union of two fuzzy sets A and B, is the set C with the same universe as that of A and B such that, the degree of an element of C is equal to the **MAXIMUM** of degrees of the element, in the two fuzzy sets.

(if Universe A $\neq$ Universe B, then take Universe C as the union of the universe A and universe B)

The **Intersection C** of two fuzzy sets A and B is the fuzzy set in which, the degree of an element of C is equal to the **MINIMUM** of degrees in the two fuzzy sets.

Example:

$A = \{\text{Mohan}/.85; \text{Sohan}/.4; \text{John}/.6; \text{Abdul}/1; \text{Abrahm}/0\}$

$B = \{\text{Mohan}/.75; \text{Sohan}/.6; \text{John}/0; \text{Abdul}/.8; \text{Abrahm}/.3\}$

Then

$A \cup B = \{\text{Mohan}/.85; \text{Sohan}/.6; \text{John}/.6; \text{Abdul}/1; \text{Abrahm} /.3\}$

$A \cap B = \{\text{Mohan}/.75; \text{Sohan}/.4; \text{John}/0; \text{Abdul}/.8; \text{Abrahm}/0\}$

and, the complement of A denoted by A' is given by

$C' = \{\text{Mohan}/.15; \text{Sohan}/.6; \text{John}/.4; \text{Abdul}/0; \text{Abrahm} /1\}$

Properties of Union, Intersection and Complement of Fuzzy Sets:

The following properties which hold for ordinary sets, also, hold for fuzzy sets

Commutativity

$$(i) A \cup B = B \cup A$$

$$(ii) A \cap B = B \cap A$$

We prove only (i) above just to explain, how the involved equalities, may be proved in general.

Let $U = \{x_1, x_2, \dots, x_n\}$. be universe for fuzzy sets A and B

If $y \in A \cup B$, then y is of the form $\{x_i/d_i\}$ for some i

$y \in A \cup B \Rightarrow y = \{x_i/e_i\}$ as member of A and

$y = \{x_i/f_i\}$ as member of B and

$$d_i = \max \{e_i, f_i\} = \max \{f_i, e_i\}$$

$$\Rightarrow y \in B \cup A.$$

Rest of the properties are stated without proof.

Associativity

$$(i) (A \cup B) \cup C = A \cup (B \cup C)$$

$$(ii) (A \cap B) \cap C = A \cap (B \cap C)$$

Distributivity

$$(i) A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$(ii) A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

DeMorgan's Laws

$$(A \cup B)' = A' \cap B'$$

$$(A \cap B)' = A' \cup B'$$

Involution or Double Complement

$$(A')' = A$$

Idempotence

$$A \cap A = A$$

$$A \cup A = A$$

Identity

$$A \cup U = U \quad A \cap U = A$$

$$A \cap \phi = A \quad \phi \cap A = \phi$$

where ϕ : empty fuzzy set = $\{x/0 \text{ with } x \in U\}$ and U : universe = $\{x/1 \text{ with } x \in U\}$

Check Your Progress - 2

Ex. 2: For the following fuzzy sets

$$A = \{a/.5, b/.6, c/.3, d/0, e/.9\} \text{ and}$$

$$B = \{a/.3, b/.7, c/.6, d/.3, e/.6\},$$

find the fuzzy sets $A \cap B$, $A \cup B$ and $(A \cap B)'$

Next, we discuss three operations, viz., concentration, dilation and normalization, that are relevant only to fuzzy sets and can not be discussed for (crisp) sets.

(1) Concentration of a set A is defined as

$$\text{CON}(A) = \{x/m^2A(x) | x \in U\}$$

Example:

$$\text{If } A = \{\text{Mohan}/.5; \text{Sohan}/.9; \text{John}/.7; \text{Abdul}/0; \text{Abrahm}/.2\}$$

then

$$\text{CON}(A) = \{\text{Mohan}/.25; \text{Sohan}/.81; \text{John}/.49; \text{Abdul}/0; \text{Abrahm}/.04\}.$$

In respect of concentration, it may be noted that the associated values being between 0 and 1, on squaring, become smaller. In other words, the values concentrate towards zero. This fact may be used for giving increased emphasis on a concept. If Brightness of articles is being discussed, then Very bright may be obtained in terms of

CON. (Bright).

(2) **Dilation (Opposite of Concentration)** of a fuzzy set A is defined as

$$\text{DIL}(A) = \{x/\sqrt{m_A(x)} | x \in U\}$$

Example:

$$\text{If } A = \{\text{Mohan}/.5; \text{Sohan}/.9; \text{John}/.7; \text{Abdul}/0; \text{Abrahm}/.2\}$$

then

$$\text{DIL}(A) = \{\text{Mohan}/.7; \text{Sohan}/.95; \text{John}/.84; \text{Abdul}/0; \text{Abrahm}/.45\}$$

The associated values, that are between 0 and 1, on taking square-root get increased, e.g., if the value associated with x was .01 before dilation, then the value associated with x after dilation becomes .1, i.e., ten times of the original value.

This fact may be used for decreased emphasis. For example, if colour say 'yellow' has been considered already, then 'light yellow' may be considered in terms of already discussed 'yellow' through Dilation.

(3) Normalization of a fuzzy set, is defined as $NORM(A) = \left\{ x / \left(\frac{m_A(x)}{Max} \right) \mid x \in U \right\}$.

$NORM(A)$ is a fuzzy set in which membership values are obtained by dividing values of the membership function of A by the maximum membership function.

The resulting fuzzy set, called the **normal**, (or **normalized**) **fuzzy set**, has the maximum of membership function value of 1.

Example:

If $A = \{\text{Mohan}/.5; \text{Sohan}/.9; \text{John}/.7; \text{Abdul}/0; \text{Abrahm}/.2\}$

$Norm(A) = \{\text{Mohan}/(.5 \div .9 = .55.); \text{Sohan}/1; \text{John}/(.7 \div .9 = .77.); \text{Abdul}/0; \text{Abrahm}/(.2 \div .9 = .22.)\}$

Note: If one of the members has value 1, then $Norm(A) = A$,

Relation & Fuzzy Relation

We know from our earlier background in Mathematics that a relation from a set A to a set B is a subset of $A \times B$.

For example, The relation of father may be written as $\{\{\text{Dasrath}, \text{Ram}\}, \dots\}$, which is a subset of $A \times B$, where A and B are sets of persons living or dead.

The relation of Age may be written as

$\{(\text{Mohan}, 43.7), (\text{Sohan}, 25.6), \dots\}$,

where A is set of living persons and B is set of numbers denoting years.

Fuzzy Relation

In fuzzy sets, every element of the universal set occurs with some degree of membership. **A fuzzy relation may be defined in different ways.** One way of defining fuzzy relation is to assume the underlying sets as crisp sets. We will discuss only this case.

Thus, a **relation from A to B**, where we assume A and B as crisp sets, is a **fuzzy set**, in which with each element of $A \times B$ is associated a degree of membership between zero and one.

For example:

We may define the relation of UNCLE as follows:

(i) x is an UNCLE of y with degree 1 if x is brother of mother or father,

- (ii) x is an UNCLE of y with degree .7 if x is a brother of an UNCLE of y , and x is not covered above,
- (iii) x is an UNCLE of y with degree .6 if x is the son of an UNCLE of mother or father.

Now suppose

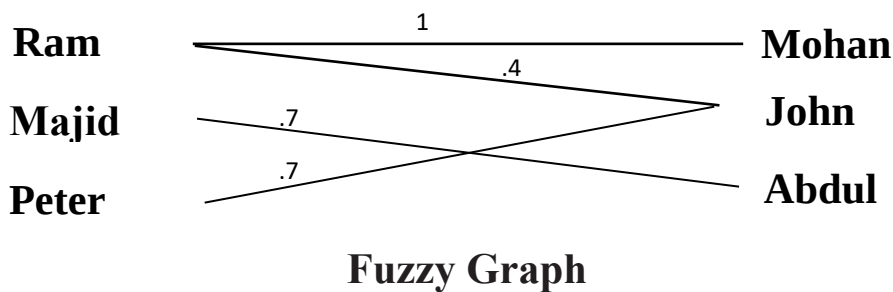
Ram is UNCLE of Mohan with degree **1**, Majid is UNCLE of Abdul with degree .7

and Peter is UNCLE of John with degree .7. Ram is UNCLE of John with degree .4

Then **the relation of UNCLE can be written** as a set of ordered-triples as follows:

$\{(Ram, Mohan, 1), (Majid, Abdul, .7), (Peter, John, .7), (Ram, John, .4)\}$.

As in the case of ordinary relations, we can use matrices and graphs to represent FUZZY relations, e.g., the relation of UNCLE discussed above, may be graphically denoted as



Fuzzy Reasoning

In the rest of this section, we just have a fleeting glance on Fuzzy Reasoning.

Let us recall the well-known Crisp Reasoning Operators

- (i) AND
- (ii) OR
- (iii) NOT
- (iv) IF P THEN Q
- (v) P IF AND ONLY IF Q

Corresponding to each of these operators, there is a fuzzy operator discussed and defined below. For this purpose, we assume that P and Q are fuzzy propositions with associated degrees, respectively, $\deg(P)$ and $\deg(Q)$ between 0 and 1.

The $\deg(P) = 0$ denotes P is False and $\deg(P) = 1$ denotes P is True.

Then the operators are defined as follows:

- (i) **Fuzzy AND to be denoted by $\wedge$, is defined as follows:**

For given fuzzy propositions P and Q, the expression $P \wedge Q$ denotes a fuzzy proposition with $\text{Deg}(P \wedge Q) = \min(\text{deg}(P), \text{deg}(Q))$

Example: Let P: Mohan is tall with $\text{deg}(P) = .7$

Q: *Mohan is educated* with $\text{deg}(Q) = .4$

Then $P \wedge Q$ denotes: 'Mohan is tall and educated' with degree $((\min)\{.7, .4\}) = .4$

(ii) Fuzzy OR to be denoted by $\vee$, is defined as follows:

For given fuzzy propositions P and Q, $P \vee Q$ is a fuzzy proposition with

$\text{Deg}(P \vee Q) = \max(\text{deg}(P), \text{deg}(Q))$

Example: Let P: Mohan is tall with $\text{deg}(P) = .7$

Q: Mohan is educated with $\text{deg}(Q) = .4$

Then $P \vee Q$ denotes: '*Mohan is tall or educated*' with degree $((\max)\{.7, .4\}) = .7$

8.5 FUZZY REASONING

The Fuzzy Reasoning is taken care by the following systems in general:

- 1) Non Monotonic reasoning Systems
- 2) Default Reasoning Systems
- 3) Closed World Assumption Systems

Let's start our discussion with the understanding of Non Monotonic Reasoning Systems

1) NON-MONOTONIC REASONING SYSTEMS

Monotonic Reasoning: The conclusion drawn in PL and FOPL are only through (valid) deductive methods. When some axiom is added to a PL or an FOPL system, then, through deduction, we can draw more conclusions. Hence, more additional facts become available in the knowledge base with the addition of each axiom. Adding of axioms to the knowledge base increases the amount of knowledge contained in the knowledge base. Therefore, the set of facts through inferences in such systems **can only grow larger** with addition of each axiomatic fact. Adding of new facts can not reduce the size of K.B. Thus, amount of knowledge **monotonically** increases with the number of independent premises due to new facts that become available.

However, in everyday life, many times in the light of new facts that become available, we may have to revise our earlier knowledge. For example, we consider a sort of deductive argument in FOPL:

- (i) Every bird can fly long distances
- (ii) Every pigeon is a bird. (iii) Tweety is a pigeon.

Therefore, Tweety can fly long distances.

However, later on, we come to know that Tweety is actually a hen and a hen cannot fly long distances. Therefore, we have to revise our belief that Tweety can fly over long distances.

This type of situation is not handled by any monotonic reasoning system including PL and FOPL. This is appropriately handled by Non-Monotonic Reasoning Systems, which are discussed next.

A **non-monotonic reasoning system** is one which allows retracting of **old knowledge due to discovery of new facts** which contradict or invalidate a part of the current knowledge base. Such systems also take care that retracting of a fact may necessitate a chain of retractions from the knowledge base or even reintroduction of earlier retracted ones from K.B. Thus, *chain-shrink* and *chain emphasis* of a K.B and reintroduction of earlier retracted ones are part of a non-monotonic reasoning system.

To meet the requirement for reasoning in the real-world, we need non-monotonic reasoning systems also, in addition to the monotonic ones. This is true specially, in view of the fact that it is not reasonable to expect that **all the knowledge needed** for a set of tasks could be acquired, validated, and loaded into the system just at the outset. In general, initial knowledge is an *incomplete* set of *partially true* facts. The set may also be redundant and may contain inconsistencies and other sources of uncertainty.

Major components of a Non-Monotonic reasoning system

Next, we discuss a typical non-monotonic reasoning system (**NMRS**) consists of the following three major components:

- (1) Knowledge base (KB),
- (2) Inference Engine (IE),
- (3) Truth-Maintenance System (TMS).

The **KB** contains information, facts, rules, procedures etc. relevant to the type of problems that are expected to be solved by the system. The component **IE** of NMRS gets facts from KB to draw new inferences and sends the new facts discovered by it (i.e., IE) to KB. The component TMS, after addition of new facts to KB, either from the environment or through the user or through IE, checks for validity of the KB. It may happen that the new fact from the environment or inferred by the IE may conflict/contradict some of the facts already in the KB. In other words, an *inconsistency* may arise. In case of inconsistencies, TMS retracts some facts from KB. Also, it may *lead to a chain of retractions* which may require interactions between KB and TMS. Also, some new fact either from the environment or from IE, may invalidate some earlier retractions requiring reintroduction of earlier retracted facts. This may lead to a chain of reintroductions. These retrievals and introductions are taken care of by TMS. The IE is completely relieved of this responsibility. Main job of IE is to conclude new facts when it is supplied a set of facts.

Let us assume KB has two facts P and $\sim Q \rightarrow \sim P$ and a rule called Modus Tollens. When **IE** is supplied these knowledge items, it concludes Q and sends Q to KB. However, through interaction with the environment, **KB** is later supplied with the information that $\sim P$ is more appropriate than P. Then **TMS**, on the addition of $\sim P$ to KB, finds that KB is no more consistent, at least, with P. The knowledge that $\sim P$ is more appropriate, suggests that P be retracted. Further Q was concluded assuming P as True. But, in the new situation in which P is assumed to be not appropriate, Q also becomes inappropriate. **P and Q are not deleted from KB, but are just marked as dormant or ineffective.** This is done in view of the fact that later on, if again, it is found appropriate to include P or Q or both, then, instead of requiring some mechanism for adding P and Q, we just remove marks that made these dormant.

- 1) Revisable belief systems
- 2) Incomplete K.B. 
 - Default Reasoning
 - Closed World assumption

In the previous section, we discussed uncertainty due to beliefs (which are not necessarily facts) where beliefs are changeable. Here, we discuss another form of uncertainty that occur as a result of incompleteness of the available knowledge at a particular point of time.

Whenever, for any entity relevant to the application, information is not in the KB, then a **default value** for that type of entity, is assumed and is assigned to the entity. The default assignment is not arbitrary but is based on experiments, observations or some other rational grounds. However, the typical value for the entity is removed if some information contradictory to the assumed or default value becomes available.

$$\frac{a(x) : Mb_1(x), \dots, Mb_k(x)}{C(x)} \quad (A)$$

260

The inference rule (A) states that if $a(x)$ is true and none of the conditions $b_k(x)$ is in conflict or contradiction with the K.B, then you can deduce the statement $C(x)$

The idea of default reasoning is explained through the following example:

Suppose we have

$$(i) \quad \frac{\text{Bird}(x) : \text{Mfly}(x)}{\text{Fly}(x)}$$

(ii) Bird (twitty)

M fly (x) stands for a statement of the form 'KB does not have any statement of the form that says x does not have wings etc, because of which x may not be able to fly'. In other words, Bird (x) : M fly (x) may be taken to stand for the statement 'if x is a normal bird and if the normality of x is not contradicted by other facts and rules in the KB.' then we can assume that x can fly. Combining with Bird (Twitty), we conclude that if KB does not have any facts and rules from which, it can be inferred that Twitty can not fly, then, we can conclude that twitty can fly.

Further, suppose, KB also contains

(i) Ostrich (twitty)

(ii) Ostrich (x) $\rightarrow \sim \text{FLY}(x)$.

From these two facts in the K.B., it is concluded that Twitty being an ostrich, can not fly. In the light of this knowledge the fact that Twitty can fly has to be withdrawn. Thus, Fly (twitty) would be locked. Because, default Mfly (Twitty) is now inconsistent.

Let us consider another example:

$$\frac{\text{Adult}(x) : \text{Mdrive}(x)}{\text{Drive}(x)}$$

The above can be interpreted in the default theory as:

If a person x is an adult and in the knowledge base there is no fact (e.g., x is blind, or x has both of his/her hands cut in an accident etc) which tells us something making x incapable of driving, then x can drive, is assumed.

3) CLOSED WORLD ASSUMPTION

Another mechanism of handling incompleteness of a KB is called 'Closed World Assumption' (CWA).

This mechanism is useful in applications where *most of the facts are known* and therefore it is reasonable to assume that *if a proposition cannot be proved, then it is FALSE. This is called CWA with failure as negation.*

This means if a ground atom $P(a)$ is not provable, then assume $\sim P(a)$. A predicate like LESS (x, y) becomes a ground atom when the variables x and y are replaced by constants say x by 2 and y by 3, so that we get the ground atom LESS (2, 3).

Example of an application where CWA is reasonable is that of *Airline reservation* where city-to-city flight not explicitly entered in the flight schedule or time table, *are assumed not to exist*.

AKB is **complete** if for each ground atom $P(a)$; either $P(a)$ or $\sim P(a)$ can be proved.

By the use of CWA any incomplete KB becomes complete **by the addition of the meta rule:**

If $P(a)$ can not be proved then assume $\sim P(a)$.

Example of an incomplete K.B: Let our KB contain only

(i) $P(a)$.

(ii) $P(b)$.

(iii) $P(a) \rightarrow Q(a)$.

(iv) Rule of Modus Ponens: From P and $P \rightarrow Q$, conclude Q .

The above KB is incomplete as we can not say anything about $Q(b)$ (or $\sim Q(b)$) from the given KB.

Remarks: In general, KB augmented by CWA need not be consistent i.e.,

it may contain two mutually conflicting wffs. For example, if our KB contains only $P(a) \vee Q(b)$.

(Note: from $P(a) \vee Q(b)$, we can not conclude either of $P(a)$ and $Q(b)$ with definiteness)

As neither $P(a)$ nor $Q(b)$ is provable, therefore, we add $\sim P(a)$ and $\sim Q(b)$ by using CWA.

But, then, the set of $P(a) \vee Q(b)$, $\sim P(a)$ and $\sim Q(b)$ is inconsistent.

8.6 FUZZY INFERENCE

PL and FOPL are *deductive* inferencing systems: i.e., the conclusions drawn are invariably true whenever the premises are true. However, due to limitations of these systems for making inferences, as discussed earlier, we must have other systems inferences. In addition to *Default Reasoning systems* and *Closed World Assumption systems*, we have the following useful reasoning systems:

1) **Abductive inference** System, which is based on the use of causal knowledge to explain and justify a (possibly invalid) conclusion.

Abduction Rule $(P \rightarrow Q, Q) / P$

Note that *abductive inference rule* is different from *Modus Ponens inference rule* in that in abductive inference rule, the consequent of $P \rightarrow Q$, i.e., Q is assumed to be given as True and the *antecedent* of $P \rightarrow Q$, i.e., P is inferred.

The abductive inference is useful in diagnostic applications. For example while diagnosing a disease (say P), the doctor asks for the symptoms (say Q). Also,

the doctor knows that for given the disease, say, Malaria (P); the symptoms include high fever starting with feeling of cold etc. (Q)

i.e., doctor knows $P \rightarrow Q$

The doctor then attempts to diagnose the disease (i.e., P) from symptoms. However, it should be noted that the conclusion of the disease from the symptoms may not always be correct. In general, abductive reasoning leads to correct conclusions, but the conclusions may be incorrect also. In other words, Abductive reasoning is not a **valid form** of reasoning.

Inductive Reasoning is a method of generalisation from a finite number of instances.

The rule, generally, denoted as, $\frac{P(a_1), P(a_2), \dots, P(a_n)}{(x) P(x)}$ states that from n

instances $P(a_i)$ of a predicate/property $P(x)$, we infer that $P(x)$ is True for all x .

Thus, from a finite number of observations about some property of objects, we generalize, i.e., make a general statement about all the elements of the domain in respect of the property.

For example, we may, conclude that: *all cows are white*, after observing a large number of white cows. However, this conclusion may have some exception in the sense that we may come across a black cow also. Inductive Reasoning like Abductive Reasoning, Closed World Assumption Reasoning and Default Reasoning is not *irrefutable*. In other words, these reasoning rules lead to conclusions, which may be True, but not necessarily always.

However, all the rules discussed under Propositional Logic (PL) and FOPL, including Modus Ponens etc are deductive i.e., lead to irrefutable conclusions.

8.7 ROUGH SET THEORY

Rough set theory can be regarded as a new mathematical tool for imperfect data analysis. The theory has found applications in many domains, such as decision support, engineering, environment, banking, medicine and others. It is a mechanism to deal with imprecise/imprecise knowledge, dealing with such a kind of knowledge is particularly area of research for the scientists, working in the field of Artificial Intelligence. There are various approaches to handle the imprecise knowledge, the most successful one is that of the Fuzzy logic, which was proposed by L.Zadeh, we discussed the same in our earlier sections of this unit.

In this section we will try to understand the Rough set theory approach, to manage the imprecise knowledge, it was proposed by Z. Pawlak. This theory is quite comprehensive and may be dealt as an independent discipline. It is quite connected with other theories and hence connected with various fields like AI, Machine Learning, Cognitive sciences, data mining, pattern recognition etc.

Rough set theory is quite comprehensive because of the following reasons :

- It requires no preliminary/additional information about the data as if it is the requirement of probability in statistics, or membership grades in the fuzzy set theory.

Artificial Intelligence- Knowledge Representation

- Facilitates the user with efficient tools and techniques to detect the hidden patterns
- Promotes data reductionality i.e. it reduces the original data and, find minimal datasets from the data with the similar knowledge as it is in the original dataset.
- Helps to evaluate the data significance.
- Supports the mechanism to Sets the decision rules from the data, automatically
- It is easy to understand, best suited for concurrent or parallel or distributed processing , and offers straightforward interpretation of obtained results.

Following are the basic/elementary concepts of the Rough set theory :

- 1) Some information (data, knowledge) is associated with every object of the universe of discourse
- 2) Objects characterized by the same information are indiscernible or similar in view of the available information about them. The indiscernibility relation generated in this way is the mathematical basis of rough set theory. Any set of all indiscernible (similar) objects is called an elementary set, and forms a basic granule (atom) of knowledge about the universe.
- 3) Any union of some elementary sets is referred to as a crisp (precise) set – otherwise the set is rough (imprecise, vague).
- 4) Each rough set has boundary-line cases, i.e., objects which cannot be with certainty classified, by employing the available knowledge, as members of the set or its complement. Obviously rough sets, in contrast to precise sets, cannot be characterized in terms of information about their elements. With any rough set a pair of precise sets, called the lower and the upper approximation of the rough set, is associated.

Note: The lower approximation consists of all objects which surely belong to the set and the upper approximation contains all objects which possibly belong to the set. The difference between the upper and the lower approximation constitutes the boundary region of the rough set. Approximations are fundamental concepts of rough set theory.

- 5) Rough set based data analysis starts from a data table called a decision table, columns of which are labeled by attributes, rows – by objects of interest and entries of the table are attribute values.
- 6) Attributes of the decision table are divided into two disjoint groups called condition and decision attributes, respectively. Each row of a decision table induces a decision rule, which specifies decision (action, results, outcome, etc.) if some conditions are satisfied. If a decision rule uniquely determines decision in terms of conditions – the decision rule is certain. Otherwise the decision rule is uncertain.

Note: Decision rules are closely connected with approximations. Roughly speaking, certain decision rules describe lower approximation of decisions

in terms of conditions, whereas uncertain decision rules refer to the boundary region of decisions.

- 7) With every decision rule two conditional probabilities, called the certainty and the coverage coefficient, are associated.
 - a. The certainty coefficient expresses the conditional probability that an object belongs to the decision class specified by the decision rule, given it satisfies conditions of the rule.
 - b. The coverage coefficient gives the conditional probability of reasons for a given decision. It turns out that the certainty and coverage coefficients satisfy Bayes' theorem. That gives a new look into the interpretation of Bayes' theorem, and offers a new method data to draw conclusions from data.

8.8 SUMMARY

In this unit the Fuzzy Systems are discussed along with the Introduction to Fuzzy Sets and their Representation. Later the conceptual understanding of Fuzzy Reasoning is build, and the same is used to perform the Fuzzy Inference. The unit finally discussed the concept of Rough Set Theory, also.

8.9 SOLUTIONS/ANSWERS

Check Your Progress - 1

Ex. 1: Discuss equality and subset relationship for the following fuzzy sets defined on the Universal set $X = \{a, b, c, d, e\}$

$A = \{a/.3, b/.6, c/.4, d/0, e/.7\}$; $B = \{a/.4, b/.8, c/.9, d/.4, e/.7\}$; $C = \{a/.3, b/.7, c/.3, d/.2, e/.6\}$

SOLUTION: Both A and C are subsets of the fuzzy set B, because $\deg(x \text{ in } A) \leq \deg(x \text{ in } B)$ for all $x \in X$

Similarly $\deg(x \text{ in } C) \leq \deg(x \text{ in } B)$ for all $x \in X$

Further, A is not a subset of C, because, $\deg(c \text{ in } A) = .4 > .3 = \deg(c \text{ in } C)$

Also, C is not a subset of A, because, $\deg(b \text{ in } C) = .7 > .6 = \deg(b \text{ in } A)$

Check Your Progress - 2

Ex. 2: For the following fuzzy sets $A = \{a/.5, b/.6, c/.3, d/0, e/.9\}$ and $B = \{a/.3, b/.7, c/.6, d/.3, e/.6\}$, find the fuzzy sets $A \cap B$, $A \cup B$ and $(A \cap B)'$

Solution : $A \cap B = \{a/.3, b/.6, c/.3, d/0, e/.6\}$,

where $\deg(x \text{ in } A \cap B) = \min \{ \deg(x \text{ in } A), \deg(x \text{ in } B) \}$.

$A \cup B = \{a/.5, b/.7, c/.6, d/.3, e/.9\}$,

where $\deg(x \text{ in } A \cup B) = \max \{ \deg(x \text{ in } A), \deg(x \text{ in } B) \}$.

The fuzzy set $(A \cap B)'$ is obtained from $A \cap B$, by the rule:

$\deg(x \text{ in } (A \cap B)') = 1 - \deg(x \text{ in } A \cap B)$.

Hence

$$(A \cap B)' = \{ a/.7, b/.4, c/.7, d/1, e/.4 \}$$

8.10 FURTHER READINGS

1. Ela Kumar, “ Artificial Intelligence”, IK International Publications
2. E. Rich and K. Knight, “Artificial intelligence”, Tata Mc Graw Hill Publications
3. N.J. Nilsson, “Principles of AI”, Narosa Publ. House Publications
4. John J. Craig, “Introduction to Robotics”, Addison Wesley publication
5. D.W. Patterson, “Introduction to AI and Expert Systems" Pearson publication